Agentic Creator Marketing Infrastructure: An MCP-Enabled Platform for Brands, Agencies, Creators, and Autonomous AI Marketing Agents

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Abstract

The creator economy has grown into a significant and fast-expanding segment of global marketing expenditure, with brands increasingly allocating budgets toward influencer-led campaigns. Despite this growth, the software systems that support creator marketing have largely remained human-centric web applications, built around dashboards, manual search filters, and messaging-based negotiation. Two converging technological developments now challenge this status quo: the emergence of autonomous large-language-model (LLM) agents capable of completing multi-step tasks with limited supervision, and the adoption of open interoperability standards — most notably the Model Context Protocol (MCP) — that allow such agents to discover and invoke external tools in a structured, predictable manner. This paper presents the design, architecture, and prototype evaluation of a Creator Marketing Infrastructure (CMI) platform that is natively operable by both human users and AI marketing agents. The system exposes a governed MCP server through which an AI agent can search for creators, draft campaign briefs, negotiate and extend collaboration offers, track deliverables, and release payments, all while remaining subject to budget scoping, permission boundaries, and human-approval checkpoints for high-impact actions. A shared data and business-logic core ensures that actions performed by an agent are immediately visible, auditable, and reversible through the conventional dashboard used by brands, agencies, and creators. The paper documents the problem context, reviews related work in creator-economy platforms and agentic AI infrastructure, describes the proposed system architecture and technology stack, outlines the implementation methodology, and presents an evaluation framework comparing agent-operated workflows against traditional manual workflows on time-to-launch, operational cost, and creator-brand matching accuracy. The study concludes that a well-governed, tool-based interface can make agentic automation of creator marketing both operationally useful and safely auditable, representing a realistic near-term evolution of existing marketing-technology (MarTech) platforms.

Keywords: Creator Economy, Influencer Marketing, Agentic AI, Model Context Protocol (MCP), MarTech Infrastructure, Multi-Agent Systems, Campaign Automation

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1. Introduction

1.1 Background

Influencer and creator marketing has moved from being an experimental, campaign-by-campaign tactic to a recurring, budgeted channel within brand marketing strategy. Brands now routinely partner with creators across platforms such as Instagram, YouTube, and TikTok to reach niche, highly engaged audiences in ways that traditional advertising increasingly struggles to replicate. Alongside this shift in

marketing practice, the broader software industry has undergone two parallel transformations that are directly relevant to how creator marketing could be operated in the near future.

The first transformation is the rise of agentic artificial intelligence. Large language models have progressed from single-turn question answering to multi-step, tool-using agents capable of planning a sequence of actions, calling external systems, observing results, and adjusting their next step accordingly — all with limited or staged human supervision. The second transformation is the standardisation of how such agents connect to external software. The introduction of the Model Context Protocol (MCP) gives AI agents a uniform, descriptive way to discover the tools a piece of software exposes and to invoke them safely, in much the same way that REST APIs once standardised communication between web services. Taken together, these two developments make it technically feasible for routine, rules-based marketing operations to be delegated to an AI agent acting under clearly defined constraints.

1.2 Motivation

Creator marketing is particularly well suited to this kind of automation because much of the day-to-day operational work is repetitive and data-driven: identifying creators who match a campaign's audience and niche requirements, drafting and sending outreach or briefs, tracking whether agreed deliverables have been posted, and reconciling payments against completed work. Existing creator-marketing platforms were not designed with this possibility in mind. They expose graphical interfaces intended exclusively for a human operator, with no documented, structured interface through which an external AI agent could act on behalf of a brand or agency. Any attempt to automate these platforms today requires brittle, unofficial integration — screen scraping, undocumented internal APIs, or custom one-off connectors — which is difficult to build, audit, and trust with real budgets. This project is motivated by the observation that creator-marketing software has an opportunity to be rebuilt around a shared, dual interface: a conventional dashboard for human actors, and a governed, standards-based MCP server for AI agents, both operating on a single, consistent source of truth.

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2. Literature Review

2.1 Creator-Economy Platforms

Current creator-marketing and influencer-marketplace platforms largely digitise manual workflows that were previously carried out over spreadsheets and direct messages. Typical functionality includes creator discovery through search filters (audience size, niche, engagement rate), in-platform messaging for outreach and negotiation, content submission and approval workflows, and payment or payout tracking. Industry benchmark reports on influencer marketing consistently describe continued year-on-year growth in brand spend directed toward creator partnerships, alongside persistent complaints from marketers about the operational overhead of running campaigns at scale — a pattern that points to unmet automation potential within the category [2], [4], [5].

Academic and industry commentary on the creator economy has tended to focus on questions of creator compensation models, platform governance, content-moderation policy, and audience-measurement methodology. Comparatively little attention has been paid to the operational software layer that brands

and agencies actually use to run campaigns day to day, which represents a gap this project addresses directly.

2.2 Agentic AI and Tool-Use Frameworks

Research and engineering practice around AI agents has matured considerably in the direction of "tool use" — equipping a language model with a defined set of callable functions and allowing it to decide, step by step, which functions to call and with what arguments in order to complete a broader task. Foundational treatments of intelligent agents, planning, and acting under uncertainty provide the conceptual grounding for this behaviour [3]. More recent engineering work has focused on making tool-use safe and auditable in production settings: scoping what an agent is permitted to do, logging every action it takes, and inserting human checkpoints before high-impact or irreversible operations are executed. These safety patterns — rather than the underlying language-modelling techniques themselves — are the most directly relevant prior work for a system, such as the one proposed here, that intends to let an agent move real marketing budget.

2.3 The Model Context Protocol

The Model Context Protocol was introduced as an open standard that lets an AI assistant connect to external data sources and tools in a uniform, machine-readable way, replacing bespoke, one-off integrations between a given model and a given piece of software [1]. An MCP server declares a set of callable tools, each with a defined name, description, and input schema, which any MCP-compatible agent can discover and invoke without custom integration code. This is conceptually similar to how OpenAPI specifications allow any HTTP client to understand and call a REST API, but tailored to the way LLM-based agents reason about and select tools. For a domain such as marketing operations, MCP offers a natural boundary: the server can expose exactly the operations an agent should be allowed to perform (for example, searching creators or releasing an already-approved payment) while keeping direct database access, administrative functions, and anything outside that boundary unreachable.

2.4 Research Gap

Reviewing both bodies of work together shows that general-purpose agent frameworks and interoperability standards such as MCP have so far been explored primarily in generic or developer-tooling contexts, rather than applied to a concrete, operationally sensitive business domain such as marketing spend. At the same time, creator-marketing platforms remain built exclusively for human operators, with no equivalent of an agent-facing interface. No existing platform combines (a) the day-to-day operational surface of a creator-marketing system — discovery, negotiation, tracking, payment — with (b) a standardised, governed tool interface that lets an AI agent operate that surface autonomously while keeping brands, agencies, and creators informed and in control. This is the gap the present project addresses through the design and prototyping of the Creator Marketing Infrastructure (CMI) platform.

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3. Problem Statement

Brands and agencies running creator-marketing campaigns today rely on manual, fragmented workflows: creators are discovered across spreadsheets, social-platform search, and siloed marketplace tools; rates

and deliverables are negotiated informally over email or direct message; deliverables are tracked by hand against a brief; and payments are reconciled separately from performance data, often in a different system entirely. As AI agents become capable of taking over this kind of routine operational work, two related problems emerge. First, purely human-operated platforms remain inherently slow and labour-intensive for campaign types that are, in principle, automatable — simple creator shortlisting, standard-rate offers, and milestone-based payment release. Second, because no creator-marketing platform exposes a documented, structured interface for non-human operators, any AI agent that is built to automate this work must integrate with each platform's proprietary and undocumented internal interface, making agent-driven marketing automation difficult to build, maintain, and audit safely. There is accordingly a need for creator-marketing infrastructure that is natively agent-operable, fully auditable, and safe by design from the outset, rather than automated after the fact through fragile workarounds.

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4. Objectives

The project is guided by the following objectives:

1. To design a unified infrastructure architecture capable of serving four distinct actor types — brands, agencies, creators, and AI marketing agents — from a single, consistent data and business-logic core.
2. To define and implement an MCP server exposing a well-scoped set of marketing tools, including creator search, campaign-brief drafting, rate negotiation, deliverable tracking, and payment release, that AI agents can call securely.
3. To design a campaign and contract data model that maintains a single source of truth shared between the human-facing dashboard and the agent-facing MCP layer.
4. To build a working prototype covering the core creator-discovery, campaign-management, and payment-release workflows end to end.
5. To define and implement guardrails — budget caps, human-approval checkpoints, and a full action audit log — that keep AI-agent-run campaigns safe and brand-compliant.
6. To evaluate the prototype against traditional, manual creator-marketing workflows on time-to-launch, operational cost, and accuracy of creator-brand matching.

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5. Proposed System

5.1 Overview

The proposed platform, referred to here as the Creator Marketing Infrastructure (CMI), is organised around a shared core that is accessed through two parallel interfaces: a conventional web dashboard for human users, and an MCP server for AI agents. Both interfaces operate against the same underlying data and business logic, so that an action taken by an AI agent — for example, booking a creator collaboration — is immediately visible, and where appropriate reversible, through the human dashboard, and vice versa. This design choice is deliberate: it avoids the common failure mode of agent-automation projects in which the automated path and the human-operated path drift out of sync, each holding a different, inconsistent view of the same campaign.

5.2 System Actors

- **Brands** create campaigns, set budgets and creative briefs, approve creator shortlists, and review campaign performance.
- **Agencies** manage campaigns on behalf of multiple brand clients from a single, consolidated workspace, with client-level separation of budgets and reporting.
- **Creators** build a public profile and media kit, list rates and content niches, receive and respond to paid collaboration offers, and access creator-facing tools such as rate benchmarking and payout tracking.
- **AI Marketing Agents** are LLM-based agents acting on behalf of a brand or agency. They operate campaigns exclusively through the MCP server and are constrained to the budget and policy limits pre-approved for that specific agent session.

5.3 Agent Interaction via MCP

Rather than exposing raw database access, the MCP server exposes a defined, limited registry of callable tools. A brand's AI agent can, for example, call `search_creators()` with audience-size and content-niche filters to obtain a shortlist, call `draft_campaign_brief()` to generate and store a structured brief, call `send_collaboration_offer()` to propose a rate to a shortlisted creator, call `track_deliverable()` to check whether agreed content has been posted and meets brief requirements, and call `release_payment()` once a deliverable has been marked complete and approved. Every tool call made through this interface is logged against the agent's identity and session, checked against the budget and permission scope assigned to that agent, and — for high-impact actions such as finalising a payment above a configured threshold — routed through a human-approval checkpoint before it is allowed to execute. This combination of scoped tools, continuous logging, and selective human-in-the-loop approval is what keeps the system auditable while still allowing routine operations to run autonomously.

5.4 System Architecture

The architecture is organised into five layers, moving from the actor-facing interfaces down to external integrations:

1. **Actor layer** — the four actor types described above: brands, agencies, and creators interacting through a conventional web/app dashboard, and AI marketing agents interacting exclusively through the MCP server.
2. **Interface layer** — the Web/App Interface (human-facing dashboards for brands, agencies, and creators) and the MCP Server (the agent-facing interface layer), running in parallel against the same core.
3. **Infrastructure Core** — four cooperating engines: the Campaign Engine (campaign creation, budgeting, scheduling), the Creator Marketplace (discovery, filtering, profile matching), the Contracts & Payments engine (agreements, escrow, payout release), and the Performance Analytics engine (reach, engagement, and cost metrics).
4. **Shared Data Layer** — a single data layer holding creator profiles, campaign records, content and performance metrics, and a financial ledger, accessed identically by both the dashboard and the MCP server so that no data can drift between the human-operated and agent-operated paths.

5. **External Integrations** — social-platform APIs (Instagram, YouTube, TikTok) for reach and engagement data, payment gateways for payouts, and LLM providers that power the AI agents themselves.

This layered separation keeps the actor-facing surfaces (dashboard and MCP server) thin, while concentrating business logic, validation, and audit logging in the shared Infrastructure Core, which is the only layer permitted to write to the underlying data store.

5.5 Data Model Overview

The shared data layer is organised around four central record types that are referenced consistently across both interfaces: **Creator Profiles** (identity, niches, rate card, historical performance), **Campaigns** (brief, budget, timeline, assigned brand or agency, and status), **Collaborations** (a specific offer between a campaign and a creator, including agreed rate, deliverables, and approval state), and **Ledger Entries** (every budget allocation, escrow hold, and payout release, each traceable back to the collaboration and the actor — human or agent — that triggered it). Keeping these four record types at the centre of the data model, rather than scattering campaign and payment state across separate systems, is what allows a single audit trail to describe the full lifecycle of a campaign regardless of whether it was run manually or by an AI agent.

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6. Proposed Technology Stack

Layer	Technology (indicative)
Frontend (brand / agency / creator dashboards)	React.js, TypeScript, Tailwind CSS
Backend / core APIs	Node.js or Python (FastAPI), REST + GraphQL
Agent interface	MCP server (Model Context Protocol) exposing campaign, creator, and payment tools
AI agent layer	LLM-based agent (e.g., Claude) orchestrating tool calls via MCP
Database	PostgreSQL (relational data), Redis (caching / sessions)
Payments	Payment-gateway integration (e.g., Stripe / Razorpay) with escrow-style holds
Analytics	Event pipeline (e.g., Kafka) with a dashboarding layer (e.g., Metabase)
Hosting / DevOps	Docker containers on a cloud provider (AWS / GCP / Azure), CI/CD pipeline

The stack is deliberately conventional outside of the agent-interface layer: the frontend, backend, database, and payment-integration choices mirror standard practice for a modern SaaS marketplace application. The distinguishing element is the addition of a dedicated MCP server sitting alongside the REST/GraphQL API, sharing the same backend services and data layer rather than duplicating business logic.

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7. Module Description

7.1 Brand & Agency Module

Handles campaign creation, budget allocation, creator-shortlist approval, a multi-client workspace for agencies managing several brands, and consolidated cross-campaign reporting.

7.2 Creator Module

Provides the creator-facing profile and media kit, rate-card management, a collaboration inbox for incoming offers, content submission against a brief, and payout tracking.

7.3 AI Agent / MCP Module

Maintains the tool registry (`search_creators`, `draft_campaign_brief`, `send_collaboration_offer`, `track_deliverable`, `release_payment`), enforces permission and budget scoping on a per-agent basis, and writes a complete, immutable action audit log for every tool call.

7.4 Contracts & Payments Module

Manages digital collaboration agreements, milestone-based escrow holds, and automated payout release once a deliverable has been marked complete and approved — whether that approval was given by a human reviewer or, within policy limits, by an authorised agent action.

7.5 Analytics Module

Provides campaign-level and creator-level performance dashboards (reach, engagement, cost-per-engagement) that are consumed identically by human users deciding their next move and by AI agents deciding their next tool call.

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8. Methodology and Work Plan

The project follows an iterative, prototype-driven methodology, moving from requirement study through to evaluation across six phases.

Phase	Activity	Indicative Duration
1	Requirement study, literature survey, scope finalisation	2 weeks
2	Data-model and system-architecture design	2 weeks

Phase	Activity	Indicative Duration
3	Core module development (campaigns, marketplace, contracts)	4 weeks
4	MCP server design and tool implementation	3 weeks
5	AI-agent integration and end-to-end testing	2 weeks
6	Evaluation, documentation, and report writing	2 weeks

Each phase produces an artefact that feeds the next: Phase 1 produces a scoped requirement document; Phase 2 produces the entity-relationship model and architecture diagram described in Section 5; Phase 3 produces the working human-facing dashboard and core backend services; Phase 4 produces the MCP tool registry with request validation and audit logging; Phase 5 integrates a sample LLM-based agent against the live MCP server and exercises the full campaign lifecycle end to end; Phase 6 consolidates results against the evaluation metrics defined in Section 10 and produces the final documentation.

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9. Guardrails and Governance

Because AI marketing agents in this system are able to initiate real spend, governance is treated as a first-class design concern rather than an afterthought. Four mechanisms work together to constrain agent behaviour:

- **Budget scoping.** Every agent session is issued a maximum spend ceiling and a per-collaboration rate ceiling at the time it is authorised by the brand or agency. No tool call that would exceed either ceiling is permitted to execute.
- **Permission scoping.** Each agent is granted access only to the specific tools it needs for its assigned task (for example, a discovery-only agent may call `search_creators()` but not `release_payment()`).
- **Human-approval checkpoints.** Actions classified as high-impact — payments above a configurable threshold, offers to creators outside a pre-approved rate band, or any campaign-brief change after creator acceptance — are held in a pending state until a human reviewer on the brand or agency side approves them.
- **Immutable audit logging.** Every tool call, whether it proceeds automatically or is held for approval, is written to an append-only audit log capturing the acting agent's identity, the tool called, the arguments supplied, the outcome, and the timestamp. This log is visible to the human dashboard in real time and forms the basis of the accountability model: brands and agencies retain a complete, chronological record of every action an agent has taken on their behalf.

Together these mechanisms are intended to let routine, low-risk operations proceed autonomously while keeping meaningful financial and brand-safety decisions under human control.

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10. Evaluation Methodology

The prototype is evaluated by running a matched set of campaign scenarios through two parallel workflows: a manual workflow operated entirely by a human user through the dashboard, and an agent-operated workflow in which a sample LLM-based agent performs the same campaign through the MCP server, with human approval checkpoints enabled as described in Section 9. Three metrics are used for comparison:

1. **Time-to-launch** — the elapsed time from campaign brief creation to the first collaboration offer being sent to a creator.
2. **Operational cost** — an estimate of the human labour time required per campaign, used as a proxy for the operational overhead each workflow imposes.
3. **Creator-brand matching accuracy** — the proportion of agent-shortlisted or human-shortlisted creators that are ultimately approved by the brand reviewer, used as a proxy for shortlist quality.

Each scenario is run multiple times to account for variance in creator availability and response time, and results are reported as averages with the underlying audit log retained for each run so that any individual agent decision can be traced and reviewed after the fact.

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11. Expected Results and Discussion

The prototype is expected to demonstrate that, for well-specified campaign types — those with a clear niche, a standard rate band, and a small number of deliverables — the agent-operated workflow can complete the discovery-to-offer sequence substantially faster than the manual workflow, since the agent can evaluate many candidate creators against the brief's filters in parallel rather than sequentially. Operational cost, measured as human labour time, is expected to fall correspondingly, since the human reviewer's role narrows to approving a shortlist and clearing high-impact checkpoints rather than performing the discovery and outreach work directly. Matching accuracy is expected to be comparable between the two workflows for straightforward briefs, with the manual workflow retaining an advantage for campaigns that depend on subjective creative judgement the brief does not fully capture — for example, matching on tone or visual style rather than audience statistics. This pattern, if confirmed, would support the view that agent-operated infrastructure is most valuable as a complement to human marketers on well-structured, repetitive campaign work, rather than as a wholesale replacement for human judgement across all campaign types.

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12. Limitations

Several limitations should be acknowledged. The prototype's evaluation uses a bounded set of simulated campaign scenarios rather than live brand budgets, so results on time and cost should be read as indicative rather than production-validated figures. The quality of agent-driven creator matching is bounded by the completeness and accuracy of the underlying creator-profile data, which in a production system would depend on reliable integration with social-platform APIs. Finally, the guardrail design in Section 9 reduces but does not eliminate the risk of an agent taking an undesired action within its

approved budget and permission scope; ongoing monitoring of agent behaviour in production would remain necessary even after deployment.

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13. Scope for Future Enhancement

- **Multi-agent negotiation**, in which a brand-side agent and a creator-side agent negotiate rates directly with one another within pre-approved policy limits, rather than a single agent negotiating against a static rate card.
- **Predictive creator-brand matching** using engagement and audience-overlap models to rank candidate creators beyond simple filter-based search.
- **Expansion of the MCP tool registry** to cover content co-creation support and automated compliance or disclosure checking (for example, verifying that sponsored-content disclosure requirements are met before a deliverable is approved).
- **Cross-platform campaign orchestration**, allowing a single agent session to coordinate a campaign across Instagram, YouTube, and TikTok simultaneously rather than per platform.

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14. Conclusion

This paper has presented the design, architecture, and evaluation plan for a Creator Marketing Infrastructure built for a market in which campaigns will increasingly be planned, negotiated, and executed by AI agents working alongside human marketers. By exposing a governed, standards-based MCP interface alongside a conventional dashboard, and by keeping both interfaces bound to a single shared data core, the proposed platform aims to make agentic automation of creator marketing both operationally useful and safely auditable. The resulting prototype and its evaluation against manual workflows are intended to demonstrate the feasibility of agent-operated marketing infrastructure as a realistic, near-term evolution of existing MarTech platforms, rather than a speculative, long-horizon capability.

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